

Multi-Agent System: Theories and Applications

Lecture 12: Lotka-Volterra Population Model

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Outline

① Single-Species Models

② Multi-Species Model

Constant vs. Logistic Growth Rate

Constant growth population model:

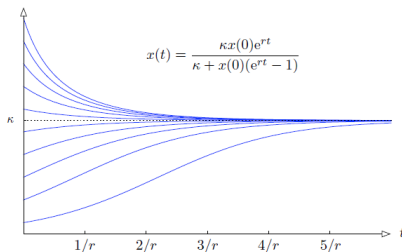
$$\frac{\dot{x}(t)}{x(t)} = r \Leftrightarrow x(t) = x(0)e^{rt}$$

→ $x(t)$ comes to either ∞ or 0 as $t \rightarrow +\infty$

Logistic growth population model:

$$\frac{\dot{x}(t)}{x(t)} = r \left(1 - \frac{x(t)}{\kappa} \right)$$

- $r > 0$: intrinsic growth rate (if $r < 0$, then $x(t) \rightarrow 0$ as $t \rightarrow +\infty$, except when $x(0) = \kappa$)
- $\kappa > 0$: carrying capacity



Lotka-Volterra multi-species population model

- Each species population is modeled by the logistic growth model
- One species influences population rates of the others *linearly*

$$\frac{\dot{x}_i(t)}{x_i(t)} = r_i \left(1 - \frac{x_i(t)}{\kappa_i} \right) + \sum_{j=1, j \neq i}^n a_{ij} x_j \Leftrightarrow \boxed{\frac{\dot{x}_i(t)}{x_i(t)} = r_i + a_{ii} x_i(t) + \sum_{j=1, j \neq i}^n a_{ij} x_j(t)}$$

where $a_{ii} = -r_i/\kappa_i$

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- Different types of inter-species influences:
 - mutualism: symbiosis and cooperation, i.e., $a_{ij} > 0$ and $a_{ji} > 0$
 - predator-prey: $a_{ij} > 0$ and $a_{ji} < 0$ for predator i and prey j
 - competition: $a_{ij} < 0$ and $a_{ji} < 0$



Taken from (Bullo, Lecture notes, 2019)

Stability analysis

• n -species Lotka-Volterra model:

$$\dot{x} = \text{diag}(x)(Ax + r) \triangleq f_{LV}(x)$$

$$A = [a_{ij}], x = [x_1, \dots, x_n]^T, r = [r_1, \dots, r_n]^T$$

• Assumptions:

(A1) A is diagonally stable, i.e., $\exists P \in \mathbb{R}^{n \times n}, P \succ 0$, and P is diagonal, such that $PA + A^T P \prec 0$

(A2) there exists a unique equilibrium point $x^* = -A^{-1}r \succ 0$

Theorem 1

If **(A1)**–**(A2)** are satisfied, then x^* is globally asymptotically stable on $\mathbb{R}_+^n$.

Proof: Hint: Consider Lyapunov function $V(x) = \sum_{i=1}^n p_i V_{\log\text{-lin}, x_i^*}(x_i)$, where p_i are diagonal elements of P , $V_{\log\text{-lin}, \kappa}(x)$ defined in Lecture 3. Then show that $L_{f_{LV}} V(x) = 1/2(x - x^*)^T (PA + A^T P)(x - x^*) \leq 0$.

Stability analysis

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Theorem 2

If $a_{ij} > 0$ for all $i \neq j$, and A is an irreducible matrix with a positive dominant eigenvalue, then all trajectories starting from $\mathbb{R}_+^n$ are unbounded, i.e., the system is unstable.

Proof: Hint: Use Perron-Frobenius Theorem, and consider Lyapunov function

$V(x) = \prod_{i=1}^n x_i^{w_i}$, where w_i are elements of the left eigenvector of A associated with the positive dominant eigenvalue.

Stability analysis

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$$A = [a_{ij}], x = [x_1, \dots, x_n]^T, r = [r_1, \dots, r_n]^T$$

Theorem 3

If $a_{ij} > 0$ for all $i \neq j$, and A is stable, then there exists a unique equilibrium point x^* which is globally asymptotically stable on $\mathbb{R}_+^n$.

Proof: Hint: Show that (A1)–(A2) are satisfied, then use Theorem 1.

Note: A matrix with non-negative off-diagonal elements is called a Metzler matrix.

Special cases for two-species model

- **2-species Lotka-Volterra model:**

$$\dot{x}_1 = x_1(r_1 + a_{11}x_1 + a_{12}x_2)$$

$$\dot{x}_2 = x_2(r_2 + a_{21}x_1 + a_{22}x_2)$$

How the behaviors of this 2-species Lotka-Volterra model for different types of interactions, i.e., mutualism, predator-prey, and competition, look like? Show explicitly.